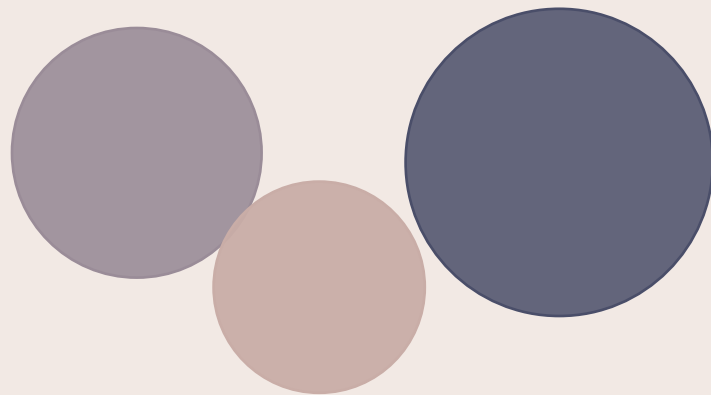


The Power of Vaccination

From Smallpox
Eradication to
COVID-19 in Record
Time

Scope: 1796–2020



Vaccines: two centuries of
innovation and impact

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Time from pathogen identification to vaccine licensure (Our World in Data).

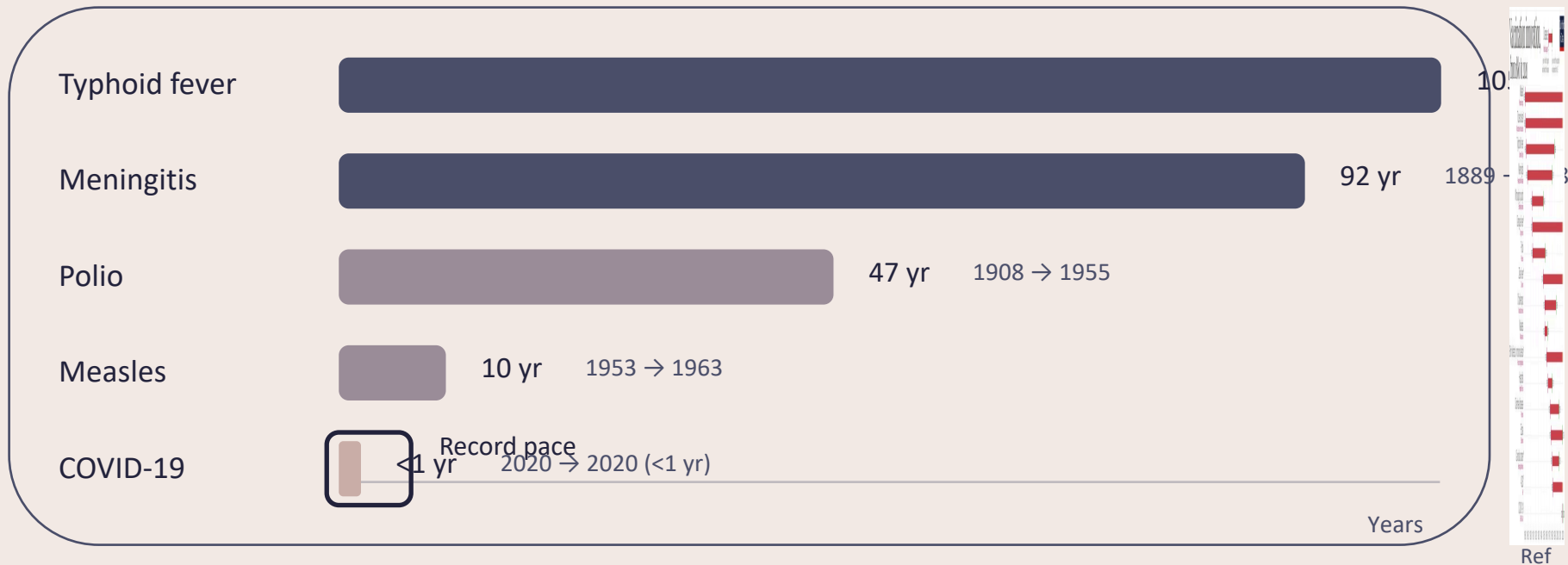


Figure reference: Vaccination innovation timeline (OWID). Data: pathogen identified → vaccine licensed.

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

US per-million annual burden before vs after vaccine introduction (Roush & Murphy, 2007; via OWID).

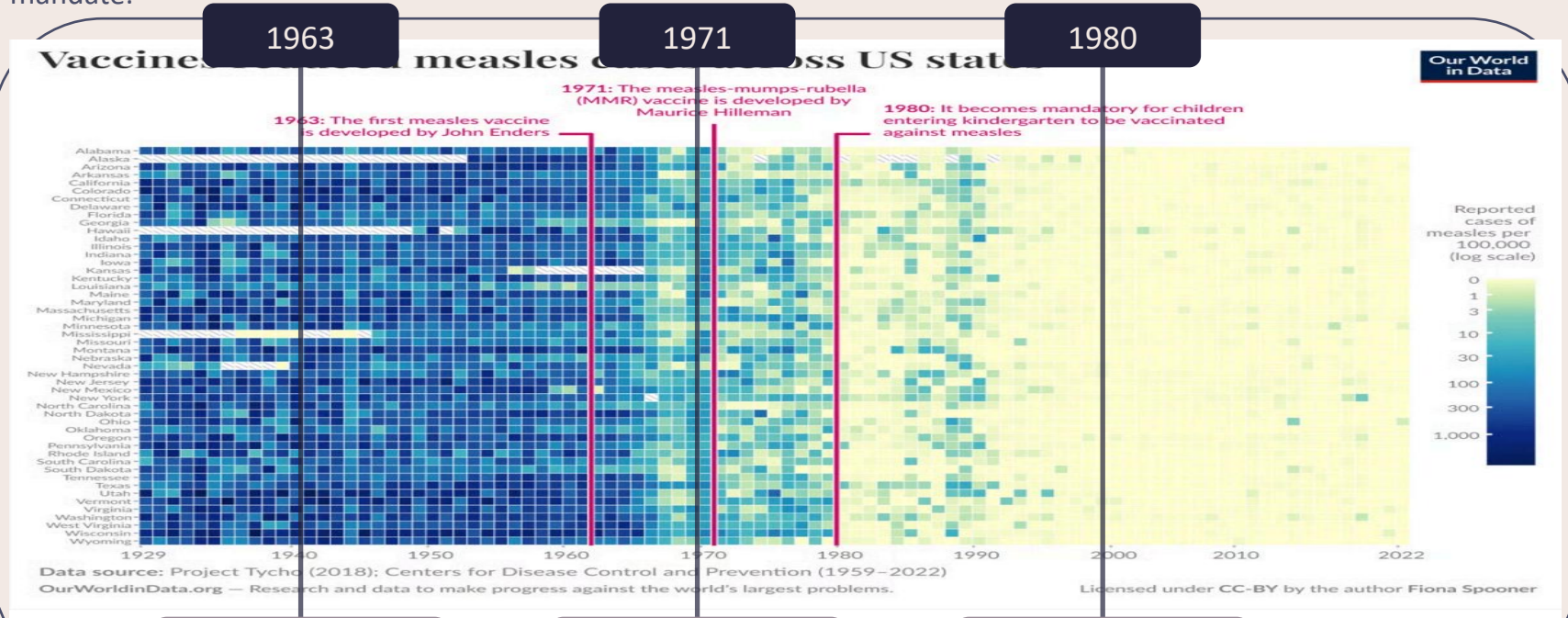
Disease	Pre-vaccine Cases /M/yr	Post-vaccine Cases /M/yr	Case reduction	Pre-vaccine Deaths /M/yr	Death reduction
Diphtheria	158	0	100%	13.7	100%
Acute polio	141	0	100%	10	100%
Smallpox	250	0	100%	2.9	100%
Measles	3,044	0.2	99.99%	2.5	100%
Pneumococcal	233	139	40.5%	24	31.3%

100% = 0 current burden in this dataset

Table values are per million population per year (US). Diseases with 100% case and death reduction are highlighted.

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

State-by-state measles cases per 100,000 (1929–2022). Milestones: 1963 vaccine, 1971 MMR, 1980 kindergarten mandate.



Interpretation: before 1963, most states saw 100–1,000+ cases per 100,000; after the 1980 mandates near zero with sporadic outbreaks (OWID).

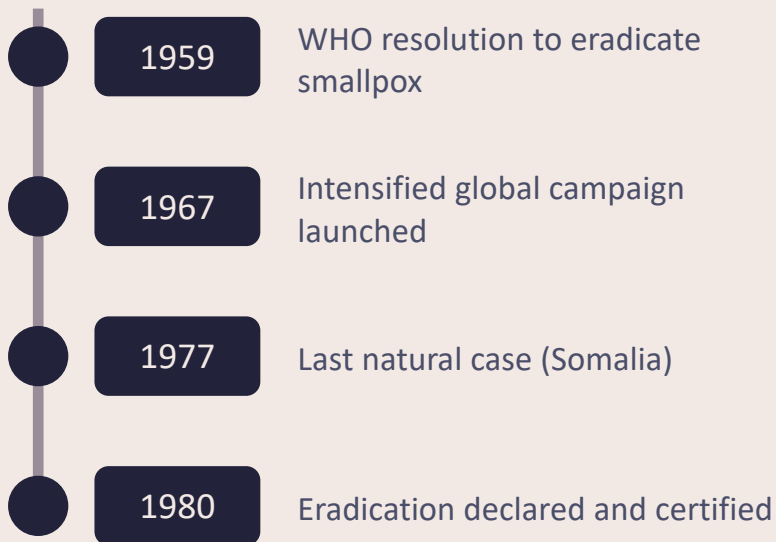
First measles vaccine (John Enders)

MMR vaccine (Maurice Hilleman)

Kindergarten mandate

Smallpox: The Only Human Disease Eradicated Through Vaccination

Eradication timeline



Why eradication was feasible: human-only infection • distinguishable symptoms • cheap, effective vaccine

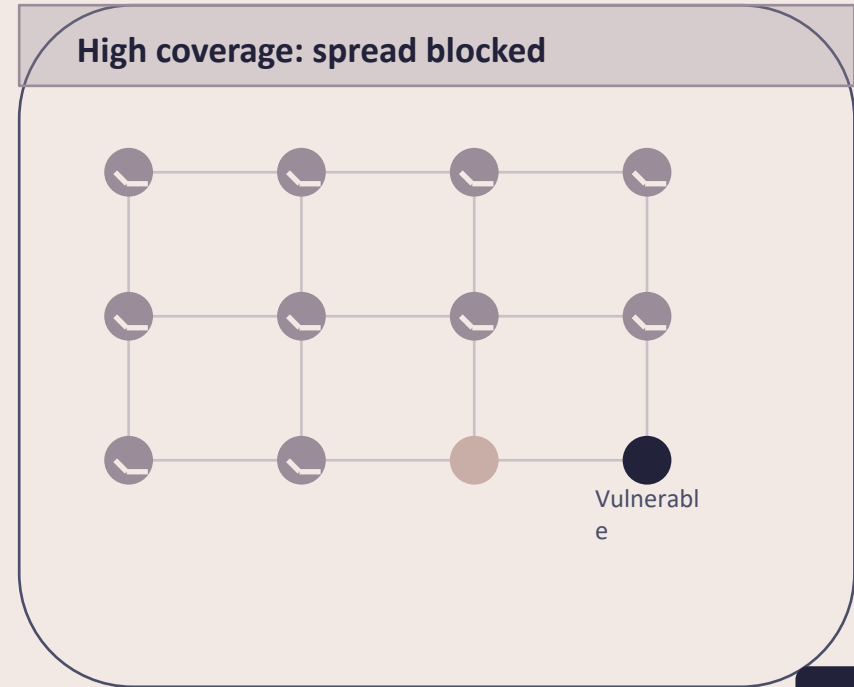
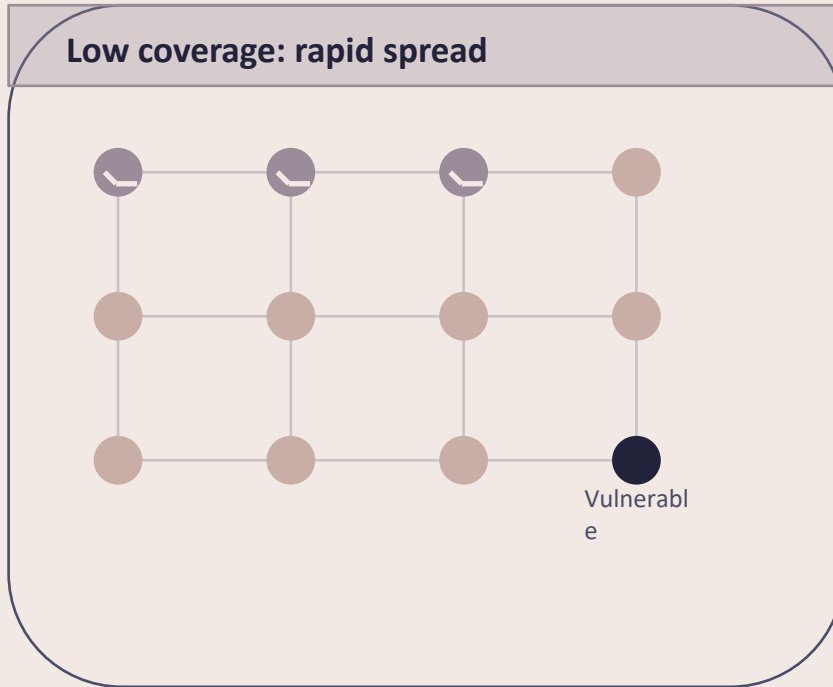
Endemic countries (reference)

	Cases		Deaths	
	All post-vaccine cases refer to 2006		All post-vaccine deaths refer to 2006	
Diphtheria	Pre-vaccine: 158 cases per million per year (1936-45)	100% reduction Post-vaccine: 0 cases per million per year	Pre-vaccine: 13.7 deaths per million per year (1936-45)	100% reduction Post-vaccine: 0 deaths per million per year
Measles	Pre-vaccine: 3044 cases per million per year (1953-62)	99.99% reduction Post-vaccine: 0.2 cases per million per year	Pre-vaccine: 2.5 deaths per million per year (1953-62)	100% reduction Post-vaccine: 0 deaths per million per year
Mumps	Pre-vaccine: 158 cases per million per year (1963-68)	97.4% reduction Post-vaccine: 22 cases per million per year	Pre-vaccine: 5.2 deaths per million per year (1963-68)	100% reduction Post-vaccine: 0 deaths per million per year
Pertussis	Pre-vaccine: 1534 cases per million per year (1941-50)	96.6% reduction Post-vaccine: 52 cases per million per year	Pre-vaccine: 30.8 deaths per million per year (1941-50)	99.7% reduction Post-vaccine: 0.09 deaths per million per year
Acute poliomyelitis	Pre-vaccine: 141 cases per million per year (1941-50)	100% reduction Post-vaccine: 0 cases per million per year	Pre-vaccine: 10 deaths per million per year (1941-50)	100% reduction Post-vaccine: 0 deaths per million per year
Paralytic poliomyelitis	Pre-vaccine: 159 cases per million per year (1951-54)	100% reduction Post-vaccine: 0 cases per million per year	Pre-vaccine: 11.8 deaths per million per year (1951-54)	100% reduction Post-vaccine: 0 deaths per million per year
Rubella	Pre-vaccine: 242 cases per million per year (1966-69)	99.98% reduction Post-vaccine: 0.04 cases per million per year	Pre-vaccine: 0.09 deaths per million per year (1966-69)	100% reduction Post-vaccine: 0 deaths per million per year
Congenital rubella syndrome	Pre-vaccine: 0.76 cases per million per year (1966-69)	99.6% reduction Post-vaccine: 0.003 cases per million per year	Pre-vaccine: no data (1966-69)	no data Post-vaccine: 0 deaths per million per year
Smallpox	Pre-vaccine: 350 cases per million per year (1900-49)	100% reduction Post-vaccine: 0 cases per million per year	Pre-vaccine: 2.0 deaths per million per year (1900-49)	100% reduction Post-vaccine: 0 deaths per million per year
Tetanus	Pre-vaccine: 4 cases per million per year (1947-49)	96.6% reduction Post-vaccine: 0.14 cases per million per year	Pre-vaccine: 3.2 deaths per million per year (1947-49)	99.6% reduction Post-vaccine: 0.01 deaths per million per year
Hepatitis A	Pre-vaccine: 465 cases per million per year (1986-95)	89% reduction Post-vaccine: 51 cases per million per year	Pre-vaccine: 0.5 deaths per million per year (1986-95)	88.7% reduction Post-vaccine: 0.06 deaths per million per year
Acute hepatitis B	Pre-vaccine: 273 cases per million per year (1982-91)	83.9% reduction Post-vaccine: 44 cases per million per year	Pre-vaccine: 1 death per million per year (1982-91)	83.6% reduction Post-vaccine: 0.16 deaths per million per year
Haemophilus influenzae type B	Pre-vaccine: 84 cases per million per year (1980)	99.8% reduction Post-vaccine: 0.17 cases per million per year	Pre-vaccine: no data (1980)	no data Post-vaccine: 0.03 deaths per million per year
Pneumococcal disease	Pre-vaccine: 233 cases per million per year (1997-99)	40.5% reduction Post-vaccine: 139 cases per million per year	Pre-vaccine: 24 deaths per million per year (1997-99)	31.3% reduction Post-vaccine: 16.5 deaths per million per year
Varicella	Pre-vaccine: 10018 cases per million per year (1990-94)	87.2% reduction Post-vaccine: 2046 cases per million per year	Pre-vaccine: 0.41 deaths per million per year (1990-94)	84.3% reduction Post-vaccine: 0.06 deaths per million per year

Smallpox eradication milestones: 1959 (WHO), 1967 (intensified campaign), 1977 (last case), 1980 (certified). Source: OWID.

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

When enough people are vaccinated, transmission chains break—shielding people who cannot be vaccinated.



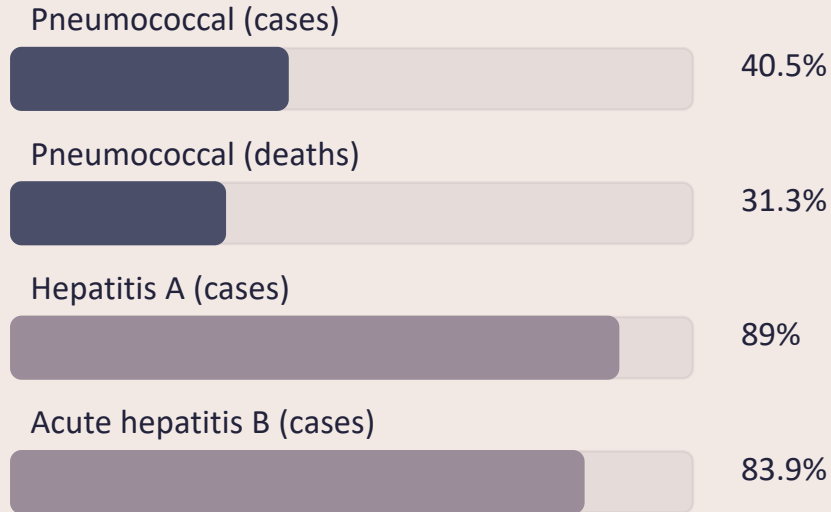
Concepts learned by community medicine effective: spread indirectly, protecting those who cannot be vaccinated (OWID brief)

Protected by community immunity: newborns • immunocompromised individuals • cancer patients on chemotherapy

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Even with major gains, some vaccine-preventable diseases show incomplete reductions and persistent access barriers.

Measured US reductions (per million/yr)



Why gaps persist

- Unequal access globally
- Supply shortages (children at risk)
- Poor scientific understanding
- Reduced trust and uptake

Policy implication:
Target coverage + access + trust

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

Elimination is possible

US data show 100% elimination of cases and deaths for diphtheria, smallpox, and acute polio.

Innovation is accelerating

Development timelines collapsed from 105 years (typhoid) to under 1 year (COVID-19).

Community protection matters

Herd immunity protects newborns, immunocompromised people, and cancer patients on chemotherapy.

Gaps persist

Incomplete reductions remain for pneumococcal disease (40.5% cases), hepatitis A (89%), and acute hepatitis B (83.9%).

Call to action: Strengthen global health systems to improve access, coverage, and trust in vaccination.

All figures and dates from the provided OWID brief (Roush & Murphy, 2007; smallpox eradication milestones; vaccine timeline table).